

AYUSH PATEL

AI Engineer · Cloud-Native & Kubernetes Expert · Open-Source Contributor · Community Leader

B.Tech, Computer Science and Engineering — Indian Institute of Technology, Delhi

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SUMMARY

AI engineer with six years of experience taking intelligent systems from research to production at global scale, spanning applied AI, cloud-native platforms and Kubernetes networking on carrier-grade infrastructure. Shipped AI experiences to millions of users, architected 5G cloud solutions and drove strategic partnerships. IIT Delhi CSE graduate; CKA & LFCS certified open-source contributor, community leader and public speaker across Korea and India.

ACADEMIC DETAILS

Year	Degree	Institute	CGPA/Percentage
2016-2020	B.Tech in Computer Science and Engineering	Indian Institute of Technology Delhi	9.040/10.0
2016	Class XII, CBSE	O. P. Jindal School, Raigarh (C.G.)	97.8%
2014	Class X, CBSE	O. P. Jindal School, Raigarh (C.G.)	10/10

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 192** in JEE Mains and **All India Rank 575** in JEE Advanced among 2 million students
- State Topper** with **97.8%** — **top 0.1%** among 1.06 million candidates in AISSCE Class XII, 2016 (CBSE)
- Selected as **KVPY Scholar** under ‘Kishore Vaigyanik Protsahan Yojana’ organised by Indian Institute of Science
- Awarded **IIT Delhi Merit Scholarship** for being in the **top 7%** among 850 students in the 1st and 2nd semester

CORPORATE EXPERIENCE/ INTERNSHIPS

AI Core Team

Software Engineer (Artificial Intelligence R&D)

Samsung Research (HQ), South Korea

Dec 2025 - Present

- Build core AI technology powering intelligent experiences across Samsung’s flagship products and services worldwide
- Take AI research from prototype to production in collaboration with research labs and business units across Samsung
- Work across the AI stack — models, data, platforms and services — delivering AI features at global product scale

SmartHome AI Lab

Software Engineer (Artificial Intelligence R&D)

Samsung Research (HQ), South Korea

Jul 2024 - Dec 2025

- Built AI experiences for millions of connected homes across the SmartThings ecosystem, deployed at a global scale
- Developed AI-driven home intelligence connecting devices, users and services seamlessly across the entire smart home

Strategic Partnership Management & Business Strategy

Solution Architect and Strategic Partnership/ Business Strategy

Samsung Networks (HQ), South Korea

Nov 2023 - Jul 2024

- Design and manage the System Integration (SI) Solution Architecture of 5G/O-RAN for various COTS, CaaS, RU/MMU
- Strategic Partnership Management (new business strategies, maintain relation, etc.) for Hyperscalers and COTS vendors

Advanced Cloud Native Solutions

Software R&D (Cloud Platform/Infrastructure Development & Operations)

Samsung Networks (HQ), South Korea

Oct 2020 - Oct 2023

- Develop Cloud Native Load Balancer for various protocols for 5G vCore & vRAN using Kubernetes Operator Framework
- Work with cloud native tools & plugins like helm, ansible, multus, cilium, DPDK, SRIOV etc. on public & private cloud
- Contribute to open-source projects like whereabouts, multus and other libraries and participate in community discussions

Founder & CTO — ChaloPadhe (Ed-Tech)

Part-time founder alongside Samsung

Association for Educational Reforms

May 2020 - Dec 2022

- Founded an ed-tech initiative making quality education accessible to students across India, leading product end to end

Tech Lead
Co-founder (Management & Operations)

Halanx Technologies Pvt. Ltd.
Jan 2020 - May 2020

- Led the Technical team comprising of App Developers, Backend Developers, Frontend Developers and UI/UX Designers
- Developed Rental Housing Solution to reduce the operating cost of owners and provide affordable housing to the tenants

Cloud Network Platform Development
Summer Internship (Cloud Software R&D)

Samsung Electronics (HQ), South Korea
May 2019 - July 2019

- Developed a Generic Toolkit for Profiling, Debugging, Automation and Monitoring Health of any Kubernetes Cluster
- Improved the available Samsung Cloud Platform (SCP) Toolkits for incorporating various CNIs (eg. OpenStack Kuryr)

DashCam Car Accident Prediction
Summer Internship (AI/ML Research Project)

ITRI, Hsinchu, Taiwan
May 2018 - July 2018

- Worked with OpenCV, scikit-learn, YOLO, Tensorflow etc. to train car accidents prediction model for self-driving cars

TECHNICAL SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Deep Learning, Computer Vision, Reinforcement Learning, OpenCV
- **Cloud Native & Platforms:** Kubernetes, Docker, Helm, Ansible, Multus, Cilium, DPDK, SR-IOV, OpenStack
- **Programming Languages:** Python, Go, C, C++, Java, C#, JavaScript, SQL, Bash, VHDL, OCaml, Scala, Prolog
- **Web & Apps:** Django, NodeJS, React, Angular, Android Studio, React-Native, Flutter, Cordova, Git, MATLAB
- **Languages Spoken:** English (professional), Hindi (native), Korean (TOPIK Level 5 — advanced working proficiency)

LICENSES & CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)** — CNCF certification for production Kubernetes clusters
- **Linux Foundation Certified System Administrator (LFCS)** — Linux operations, network and storage
- **Cloud Specialist** — Samsung Electronics internal certification for cloud platform & infrastructure specialists
- **TOPIK Level 5** (advanced Korean proficiency) · **KIIP Level 6** · **KINAT** — Korean language and integration

LEADERSHIP & COMMUNITY

- **President, IIT Alumni Association of Korea (IITAAK)** — engaging business, academia and media (2023 - Present)
- **Vice President (Public Relations), IITAAK** — grew the association's public presence across Korea (2021 - 2023)
- **Speaker at the India-Korea Forum on AI 2026**, Ewha Womans University, Seoul — on AI cooperation (July 2026)
- **Host of the Fun Cinema & VFX Booth, India Day Seoul 2026** — Embassy of India's inaugural festival (May 2026)
- **Executive Secretary, Indian Students and Researchers in Korea (ISRK)**, nationwide (2025 - Present)
- **Core Team Member, Indians in Korea (IIK)** — among Korea's largest Indian communities (2025 - Present)
- **Microsoft Student Partner India**, Delhi-NCR Region — developer workshops and campus outreach (2017 - 2019)
- **Overall Coordinator**, Nokia Data Science Hackathon; **Executive & Developer**, DevClub IITD (2017 - 2018)

MAJOR PROJECTS

Automotive Security System

B.Tech Project, IIT Delhi (2019 - 2020)

- Studied the architecture of Controller Area Network (CAN) bus and Plug-in Hybrid Electric Vehicle (PHEV) Technology
- Devised an improved architecture which tackles the existing security vulnerability without hampering the performance

Game-Bots using ML and AI

AI & ML Course Projects, IIT Delhi (2018 - 2019)

Built Robots for numerous games (eg. Yinsh, Arkanoid) using Q-learning, CNN and various Deep Learning based Models

Nokia ML Hackathon Platform

Industry Project, IIT Delhi (2018)

Developed a platform to host large-scale Data Science Hackathons using Hadoop, Docker, JupyterHub, NodeJS, Django

Baadal (IITD Computing Cloud)

Cloud Computing, IIT Delhi (2020)

Implemented Load Balancing Module, On-Demand VM migration system and Resource Monitoring dashboards in Baadal

Open-Source Kubernetes Networking (whereabouts, multus)

Network Plumbing WG (2020 - Present)

Contributed IPAM and multi-network CNI enhancements running in production Kubernetes and telco clouds worldwide